Conditional quantum measurement

Purity bounds from local output measurements

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Mathematical and computational assistance · ChatGPT (OpenAI). All probabilities in the numerical example are supplied model values.

> On one known pure product input, first-qubit tomography and one binary measurement on the second qubit give a sharp interval for channel purity. A unitary channel fits these local readings exactly when the two possible second-qubit probabilities both lie between the eigenvalues of the first-qubit state.

Let C be the normalized Choi matrix of a trace-preserving two-qubit channel. Its purity is $\mathrm{Tr}(C^2)$. On input $|00\rangle$ let Ω be the output, and let a be the larger eigenvalue of its first-qubit marginal. Write β for the probability of a specified outcome of a rank-one projective measurement on the second qubit. Set $b=\min(\beta,1-\beta)$, the less frequent population in that basis.

Exact upper endpoint

Put $m=\min(1,a+b)$. Over all such channels, the sharp upper endpoint of purity is $[9+6m+m^2+(1-m)^2]/16$. It reaches one precisely when $b\geq 1-a$. If $b<1-a$, the measured local data exclude every unitary channel. There is no need for a joint output measurement to apply this result.

Exact lower endpoint

Work in the first marginal's eigenbasis and order the second outcomes by decreasing probability. Let x be $(a+b)/2-¼$ clipped to $[\max(0,a+b-1), \min(a,b)]$. Define $L=(a-x)^2+x^2+(1-a-b+x)^2+(b-x)^2$. The sharp lower endpoint is $(L+¾)/16$. Every intermediate purity occurs. Sharpness here concerns precisely the specified local data; extra Choi observations can restrict both endpoints.

Numerical distinction

At the supplied 1% residual point, $a=0.529472934503538\ldots$. A unitary fits the local readings only if the second-qubit probability belongs to $[1-a,a]$, approximately $[0.47052707,0.52947293]$. No such second-qubit rate was measured. Page 3 uses the independently specified mathematical detector to obtain a one-sided bound on that rate.

Extremal states and channels

For a fixed joint output Ω on one pure input, put P=Tr(Ω²), λ=λmax(Ω). The exact Choi purity endpoints are (P+¾)/16 and (9+6λ+P)/16. For the minimum, the sixteen blocks of 4C are Φ(|i⟩⟨j|). Their squared Hilbert–Schmidt norms add. Send the measured input to Ω, the other three basis states to $I_4/4$ and set the coherences to zero. This attains the lower endpoint.

For the maximum, let τj be the environment state after a channel isometry on input j. The spectrum of τ_0 equals that of Ω up to zeros. The environment state on the maximally mixed input has the spectrum of C up to zeros. Writing $\tau=(\tau_1+\tau_2+\tau_3)/3$ gives 16 $\mathrm{Tr}(C^2)=P+6\mathrm{Tr}(\tau_0\tau)+9\mathrm{Tr}(\tau^2)\le P+6\lambda+9$. Choose a Schmidt purification of Ω for input 0, and map the other inputs to mutually orthogonal system vectors perpendicular to its top system Schmidt vector, all with that top environment vector. These orthonormal images attain equality.

Optimizing over unobserved joint correlations

In the measured basis of the second qubit, write Ω as a positive two-by-two block matrix with diagonal blocks A,D. Then A+D is the known first marginal and Tr D=b. Positivity bounds any cross-block expectation by the geometric mean of the two diagonal expectations. The Rayleigh quotient gives $\lambda\max(\Omega)\le\|A\|+\|D\|\le a+b$. It is also at most one. Since m≥½, the spectrum (m,1−m,0,0) majorizes every compatible spectrum, giving $P\le m^2+(1-m)^2$.

For b<1−a, these two bounds are attained together. Use $|\varphi\rangle=\sqrt{a}\,|00\rangle+\sqrt{b}\,|11\rangle$ and $\Omega=|\varphi\rangle\langle\varphi|+(1-a-b)|10\rangle\langle 10|$ in the chosen bases. For b≥1−a, a pure Schmidt state of weights a,1−a with a rotated second basis realizes the requested probability b. The two marginal spectra then coincide.

For the minimum, dephasing in both selected bases keeps the measured data and can only lower joint purity. The four diagonal probabilities are a−x, x, 1−a−b+x and b−x. Minimizing their sum of squares gives the clipped value on page 1. Convex mixtures of endpoint channels preserve the local data. Continuity of purity proves all intermediate values occur.

Related marginal compatibility results · S. Bravyi, quantum marginal inequalities.

Quantitative exclusion at the supplied model point

Normalize Choi kets as $|M\rangle=\mathrm{vec}(M)/2$. The projections onto $p=|I_4\rangle$, $u=|iZ\otimes I\rangle$ and $v=|iX\otimes I\rangle$ are orthogonal and lie in the subspace of $|A\otimes I\rangle$. Let w denote its Choi population. The measured populations of p,v are s,f. The ordinary effect is the projection onto $\sqrt{h}\,p+\sqrt{(1-h)}u$. Its probability q implies $w\geq s+f+k$ with $k=(\sqrt{q}-\sqrt{(hs)})^2/(1-h)$, by the reverse triangle inequality. Without that calibrated effect one still has $w\geq s+f$.

For a unitary U, its off-diagonal blocks that change the second bit have equal squared Frobenius norms. Their sum is bounded by $4(1-w)$. The flip probability on $|00\rangle$ is thus at most $2(1-w)$. Both marginals of its pure joint output have the same spectrum, so $a\geq 2w-1$. This reproduces the exclusion thresholds 0.749034900224 and 0.98 using two and three sector population bounds respectively, against $a\approx 0.529472934504$.

Ceiling on purity without a full phase curve

For any trace-preserving channel with Kraus matrices K_j, the sum of squared Frobenius norms of their components outside the local subspace is $4(1-w)$. Each local component acts trivially on the second bit, so the probability β of finding that output bit equal to one on input $|00\rangle$ satisfies $\beta\leq 4(1-w)$. This inequality uses neither unitality nor a pure-channel hypothesis.

> For the specified s,h,q,f the lower bound on w is 0.99. Hence $\beta\leq 0.04$ and $\mathrm{Tr}(C^2)\leq 0.807905661518$, rounded upward. This ceiling uses the ordinary effect and first-qubit tomography. Its sharpness under all the supplied Choi constraints is unresolved.

For contrast, the minimum-purity reference state of the full phase model has $\beta=8(0.01)/13$. If that rate were independently established along with the same first marginal, the interval based only on those local data would be approximately [0.07783103,0.79476870]. The full phase-model state supplies a prediction for β. Additional Choi data can further restrict that interval.

With only the five rates s,f,pX,pY,pZ in a common box of half-width 0.03 around the reference values, the unitary lower estimate is 0.629034900224 and the largest permitted a is 0.573054978875, rounded to nearest. Their difference exceeds 0.05597. The crossing of this particular box certificate lies strictly between 0.03990195042113343 and 0.03990195042113345. These box widths describe population uncertainties and carry no confidence level without an independently justified sampling procedure.

What a positive exclusion margin proves

Put $W=2(s+f)-1-a$. The submitted distance bound $W/6$ is valid. The specific projectors give a stronger certificate. For every normalized unitary Choi state C_u, its conventional half-trace distance from C satisfies $\frac{1}{2}\|C-C_u\|_1 \geq W/(3+\sqrt{3})$. At the supplied point the lower bound exceeds 0.04639890. This is a Choi-state distance. No optimality of the coefficient is asserted.

Proof

Let M project onto $\mathrm{span}\{p,v\}$. For C_u take any top eigenvector of its first output marginal on $|00\rangle$. Let Q be the projection onto its orthogonal first-qubit direction, arbitrary second output, and reference input $|00\rangle$. The unitary inequality and the variational formula for a give $W \leq \mathrm{Tr}[(2M+4Q)(C-C_u)]$. The added input-only term cancels since both Choi matrices have reference marginal $I_4/4$.

The compression MQM has eigenvalues $\frac{1}{4}$ and 0 on M. Indeed Q sends p,v to collinear vectors on the chosen first direction and second bit zero, with total squared norm $\frac{1}{4}$. In the two-dimensional principal-angle block, $2M+4Q$ has top eigenvalue $3+\sqrt{3}$. The other eigenvalues are smaller and the minimum is zero. For a traceless Hermitian difference the expectation is bounded by this spectral range times the half-trace norm. This proves the certificate.

Interpretation

The bounds constrain a set of channels. They give no probability law over compatible channels and do not identify a microscopic source of mixing. An application requires the same calibrated input and output measurements, trace preservation on the tested path, and a justified simultaneous uncertainty region. A branch selected after history 111 can have input-dependent success probability. The supplied materials contain no matched hardware records for this path.

At the interior reference state, first-qubit measurements leave 204 local real channel dimensions, or 189 with unitality imposed. The exhibited nine-dimensional family has the same full two-qubit output on $|00\rangle$. In the separate $\pm 60°$ contrast experiment, zero is attainable for general states at the exact distance cap ≈ 0.83097060; normalized trace-preserving channels have positive contrast at that cap. Their sign without the cap is unresolved.

Supplied mathematical note · Pure-channel exclusion with incomplete phase data.

A. Higuchi, A. Sudbery and J. Szulc (2003) · one-qubit marginals of pure states.

J. Watrous, The Theory of Quantum Information, ch. 2 · channel representations.

G. A. L. White et al. (2022) · experimental multitime process characterization.